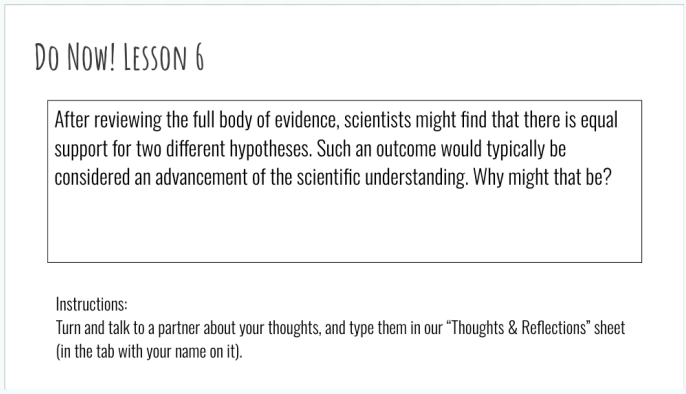


Reasoning:

During this lesson, students will synthesize findings from multiple different experiments to decide on the best mitigation recommendations for the town. Students will have to develop a conclusion that reconciles all the high-quality evidence while recognizing the social nature of this phenomenon (this helps reinforce the ideas about consensus-seeking in science discussed in Lesson 5). In this lesson, students also have the opportunity to engage in scientific communication by hosting a conference that allows students to collect high-quality evidence from their peers. The conversations in this lesson will elevate the idea that science can be uncertain and conclusions can be conditional even if multiple experiments have been conducted. By including results from multiple studies, students are conducting a mini meta-analysis.

Slides	Notes
 DO NOW! LESSON 6 After reviewing the full body of evidence, scientists might find that there is equal support for two different hypotheses. Such an outcome would typically be considered an advancement of the scientific understanding. Why might that be? Instructions: Turn and talk to a partner about your thoughts, and type them in our "Thoughts & Reflections" sheet (in the tab with your name on it).	It is important to address the idea that supporting multiple hypotheses with data is not the same as failing to resolve a disagreement (Do Now from Lesson 5). Supporting multiple hypotheses with evidence does not mean that there are different opinions. Key points to highlight: • Science is not just about proving one idea right • Evidence can support multiple explanations • Scientific knowledge is refined over time • Uncertainty can actually move science forward Students might say: • Scientists were looking at two different aspects of the same phenomenon (i.e., some scientists looked at only vaccines while others looked at masking). • Scientists used different methods to investigate their hypotheses, so they had different outcomes.
 CONFERENCE TIME 1. Each mitigation factor group will present their conclusion and evidence for their claims about their mitigation factor 2. Ask questions to make sense of all mitigation factors claims and their data.	Presentation tip for groups: focus on the last 3 conclusions of the review tab for each mitigation factor

MAKING CONCLUSIONS WITH OUR EVIDENCE

In the Conference Questions tab of your experiment report, consider the following questions:

1. What seems to be the two most effective mitigation factors? Why?
2. When are these factors effective? When are they not?
3. What is the impact of implementing this mitigation factor mandate on the population of the town? How can this impact people's personal, day to day interactions?
4. If the simulation was a real town, how might your recommendation be different?
5. What are we still unsure about? How might our conclusions be wrong?
6. What else could be done in the future to become more certain?

After students have taken some time to answer these questions in their groups, move to a class discussion of everyone's answers. If you need to save some time, it could help to skip groups answering on their own, and instead move directly to the class discussion.

Teacher discussion moves:

1. 🧑🔬 We are stopping here and identifying the 2 best mitigation factors, but usually scientists don't stop here. Some scientists might narrow down their choices of mitigation factors and then run more experiments/ collect more data to see how effective they are
2. 🌍 This simulation shows mitigation factors interacting with one another—during COVID we were told to mask and social distance if we were out in public because the interaction of the two mitigation factors was effective and supported by data.
3. 🌍, 🧑🔬 Focus on the social aspect of science. Some students might bring in examples that if virtual school/work was the best factor, but there are some people that can not do their jobs without being in person (e.g. sanitation departments, construction workers, etc.), how do people's circumstances affect their participation in of the mitigation factor?
4. 🌍 In the real world things are ever evolving. During COVID, since the virus was mutating, our science and mitigation factors had to change with it. For example, because of the omicron variant, scientists had to update the booster vaccines to target that particular variant of the epidemic.
5. 🧑🔬 Science is uncertain— as a class we collected a lot of data, but it is impossible to have all the evidence. New evidence can come to light over time and we must update our models accordingly.
6. 🔍 Here students can have a discussion about collecting more evidence, integrating evidence from other experts and studies, and planning/carrying out more controlled experiments.

END OF LESSON 6 REFLECTION

After reviewing the full body of evidence, scientists might find that there is equal support for two different hypotheses. Such an outcome would typically be considered an advancement of the scientific understanding. Why might that be?

Instructions:

Write your thoughts on this in our "Thoughts & Reflections" sheet (in the tab with your name on it).

Strong student understanding would sound like: "Scientists aren't just disagreeing—they have two explanations that both fit the evidence so far. That helps them design new experiments to figure out which one is more accurate or if a new explanation is needed."

If students say the following be sure to redirect:

- "They just have different opinions"
- "They can't decide"
- "They agree to disagree"

Possible teacher responses:

- "What role is evidence playing here?"
- "What would scientists do next?"

- | | |
|--|---|
| | <ul style="list-style-type: none">• “Does science stop when there are two ideas?” |
|--|---|